



UNIVERSITÉ DE
MONTPELLIER

Master Imagine

Codages et compressions multimédia

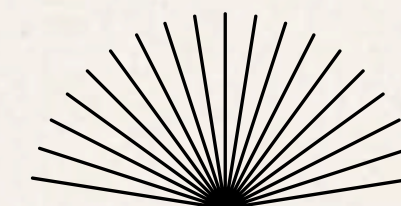
Images mosaïques

Avec utilisation de critères avancés.

CRYPTO MOSAÏC

PRÉSENTÉ PAR:

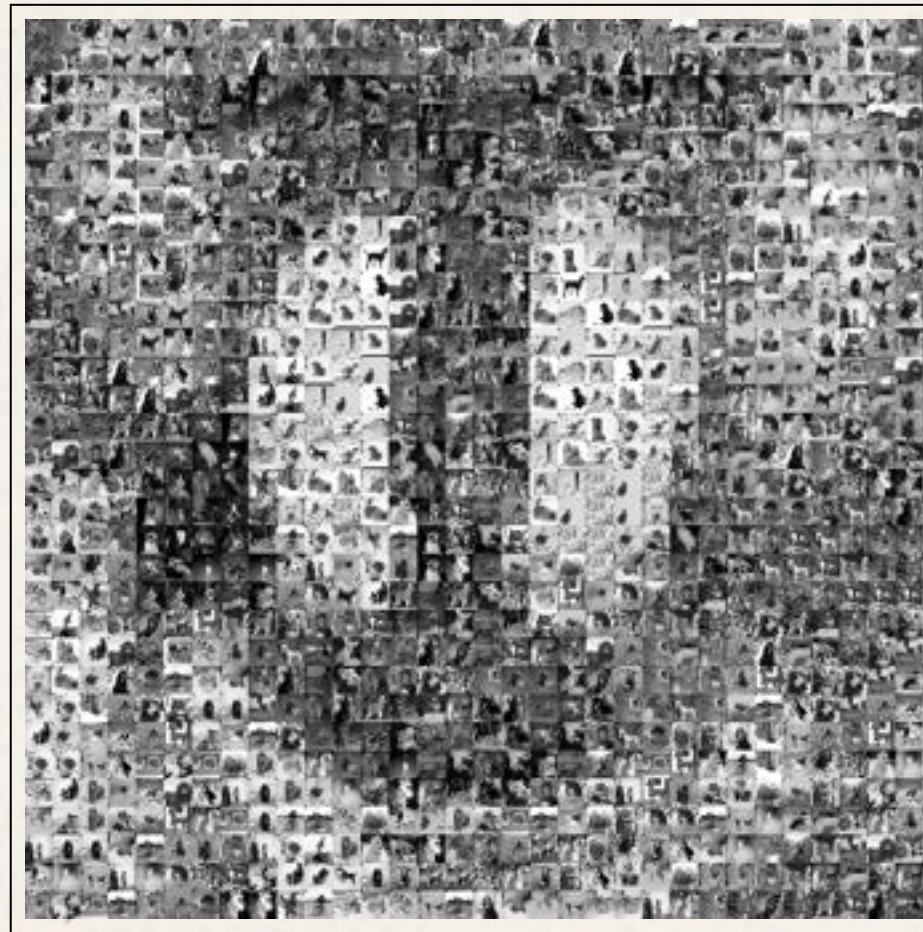
Rémy GUIBERT, Bastian LANGLOUET et Mickaël LONGLADE



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Contexte et objectifs



Contexte # 1

Codage et compression multimédia.
Analyse et traitement des images.



Objectifs # 2

Génération d'images mosaïques.
Méthodes de protections des
images.

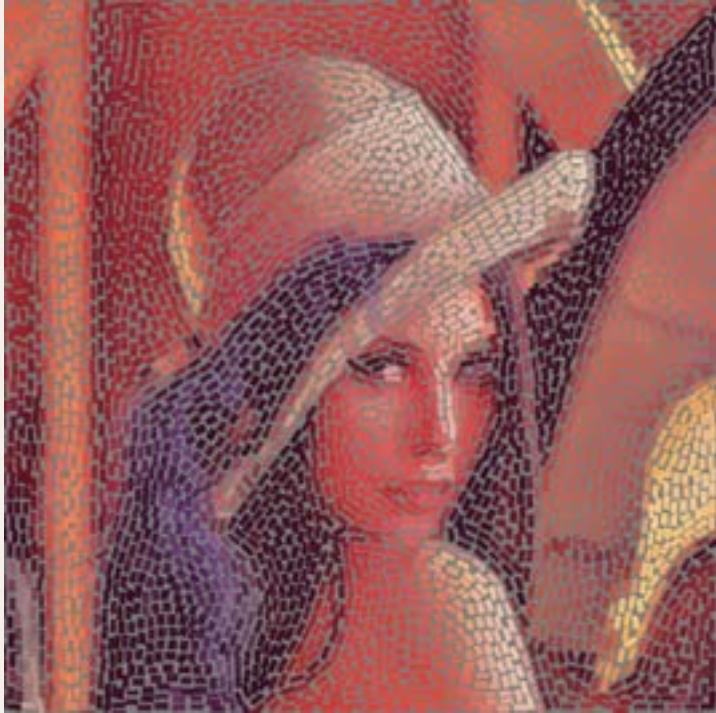


Solution # 1

Tuiles carrées avec différentes
métriques de sélection.
Chiffrement de Vernam et
brouillage par permutation.

État de l'art

GÉNÉRATION D'UNE IMAGE COMPOSÉE DE PLUSIEURS PETITES IMAGES (TUILES) CONSERVANT L'INFORMATION VISUELLE D'UNE IMAGE SOURCE.

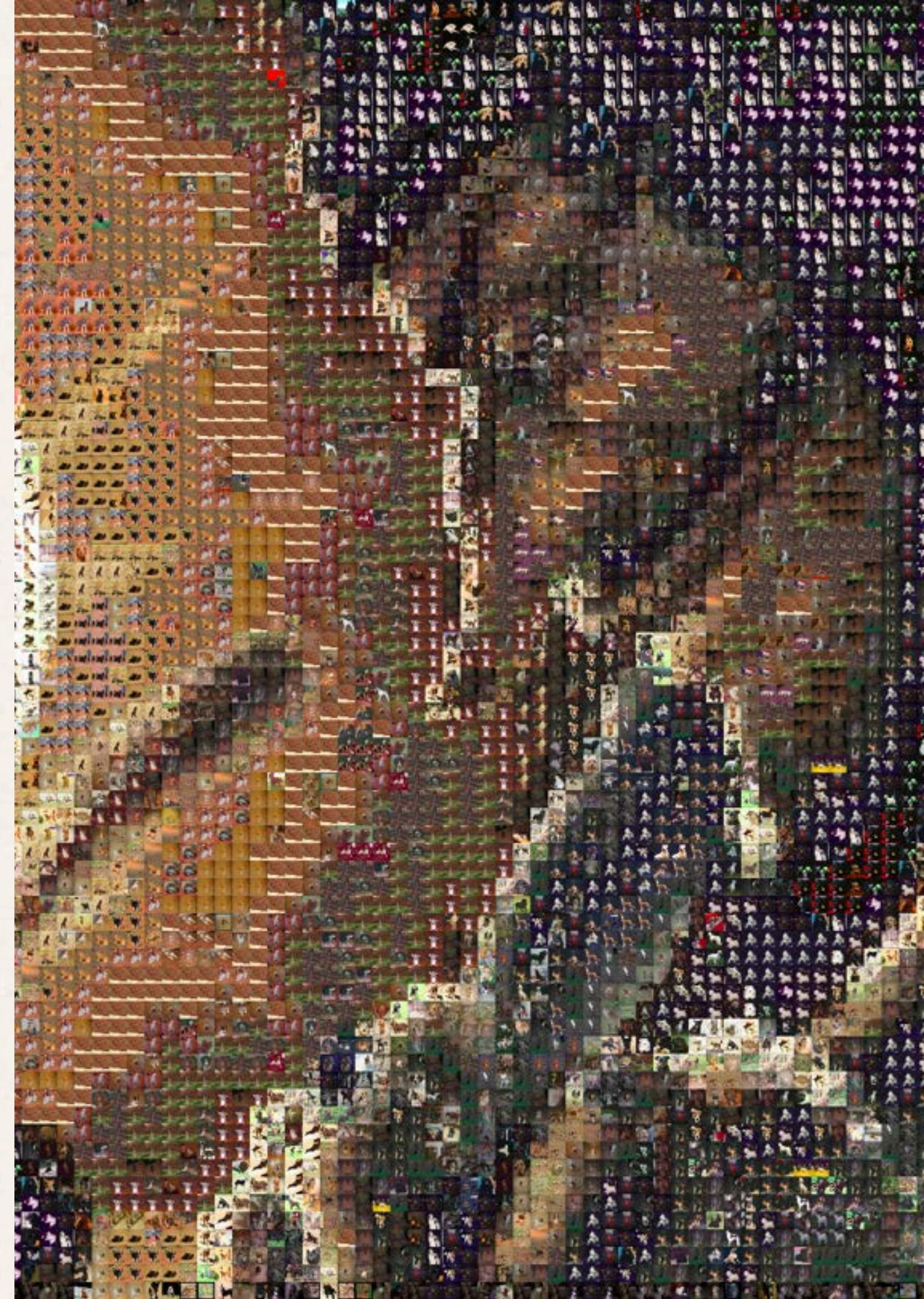


Méthodes classiques utilisant différentes métriques.
Méthodes avancées utilisant l'intelligence artificiel.
Taille des blocs.
Richesse de la base d'image et rendu.
Complexité et cout de calcul $O(W*H*T)$.

“Smart Ideas for Photomosaic Rendering”,
G. Di Blasi, G. Gallo, M.P. Petralia, 2006

Code et implémentations

- 01 Back-end en C++
- 02 Interface en Python, avec Tkinter.
- 03 Aucune utilisation de bibliothèques auxiliaires



Travaux réalisés

Images mosaïques

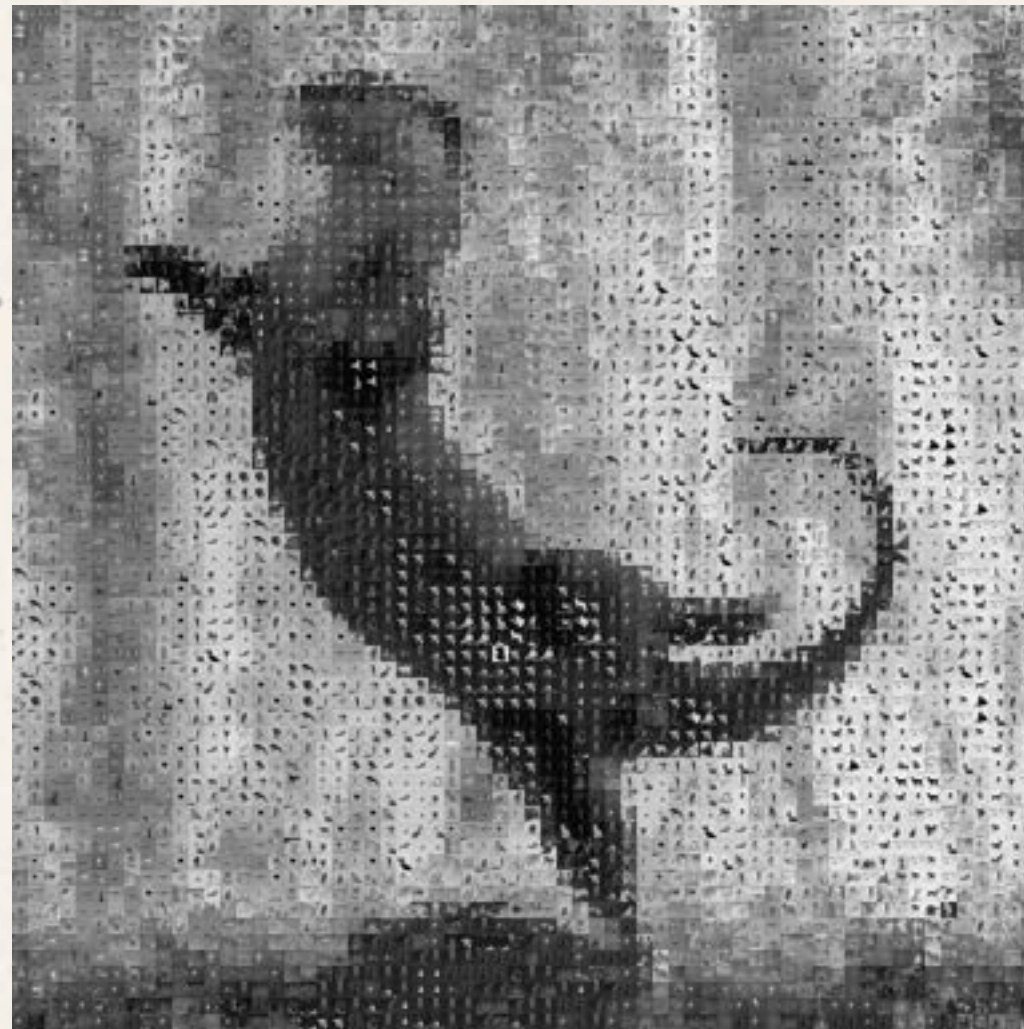


Travaux réalisés

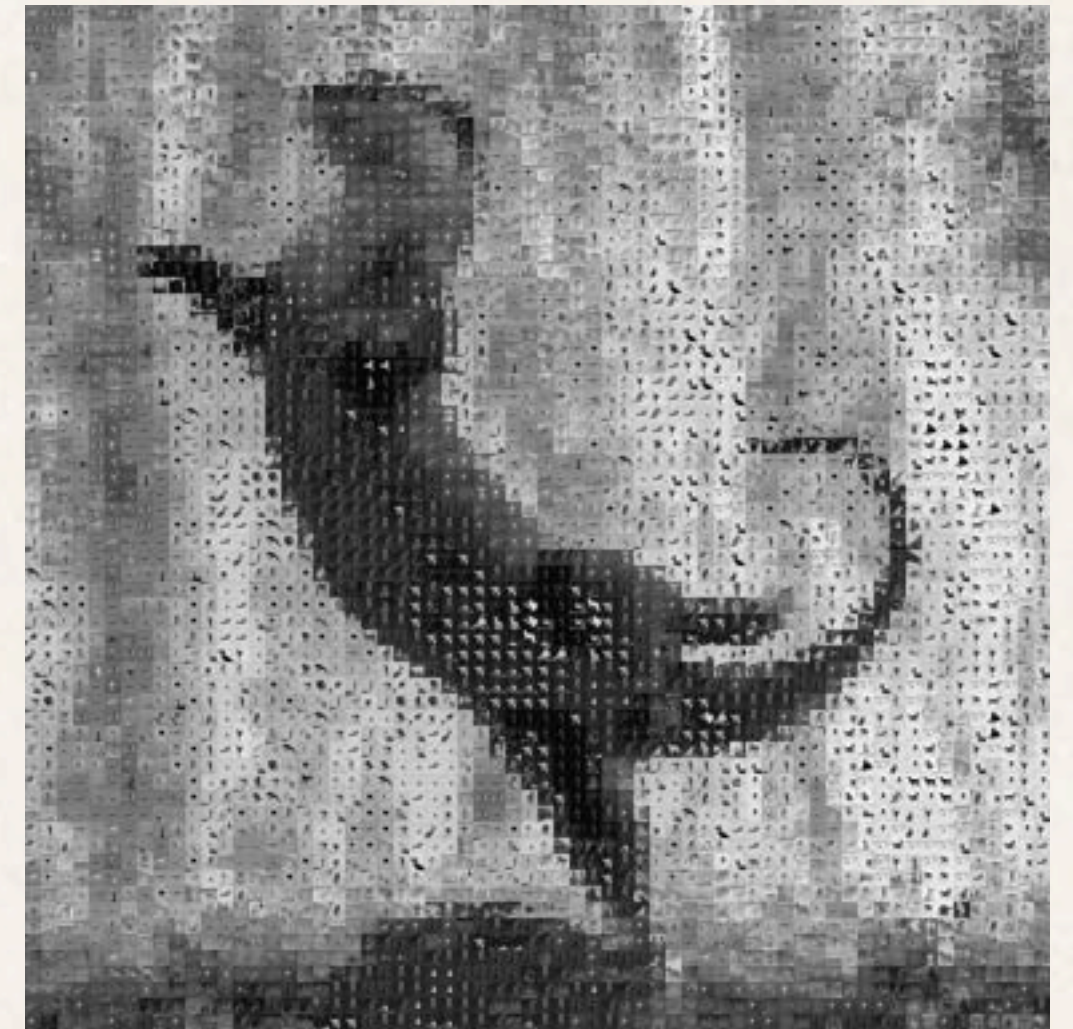
Images mosaïques



Moyenne



Bhattacharyya



Chi-2

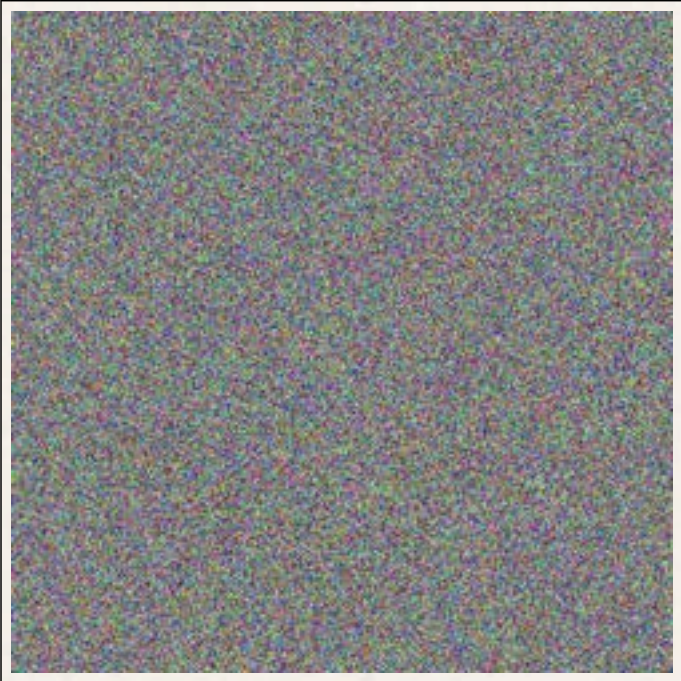
Travaux réalisés

Chiffrement et brouillage.



Loutre.ppm

Originale



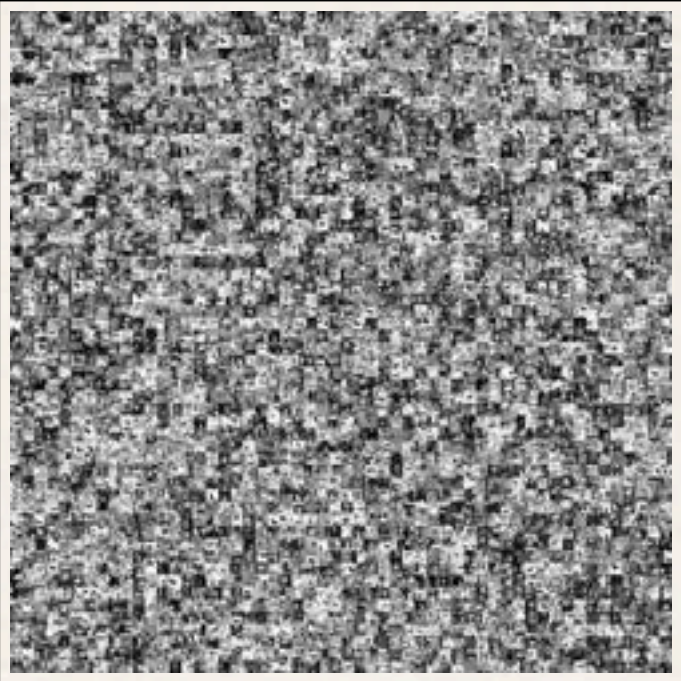
LoutreVernam.ppm

Chiffrée



LoutreMosaique.ppm

Mosaïque



LoutrePermute.ppm

Mosaïque Brouillée



Résultats

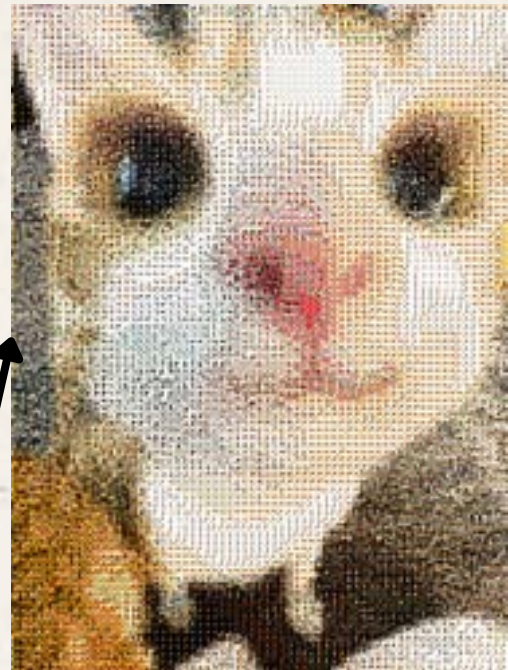
Temps de générations

Blocs de 16 x 16

1440

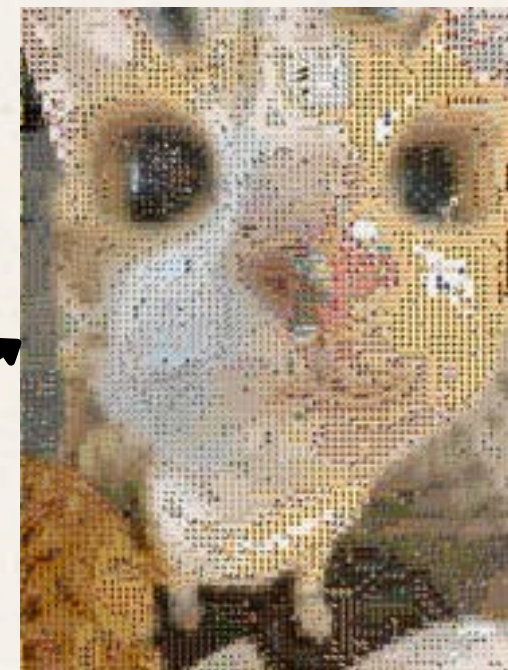


1080



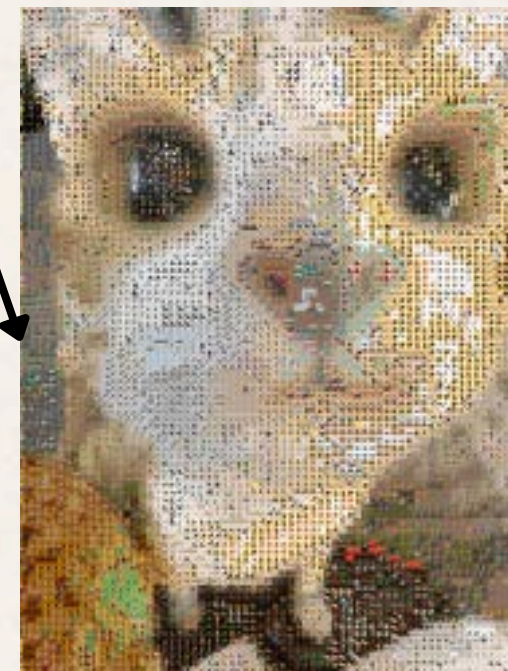
Moyenne

2 min 46



Bhattacharyya

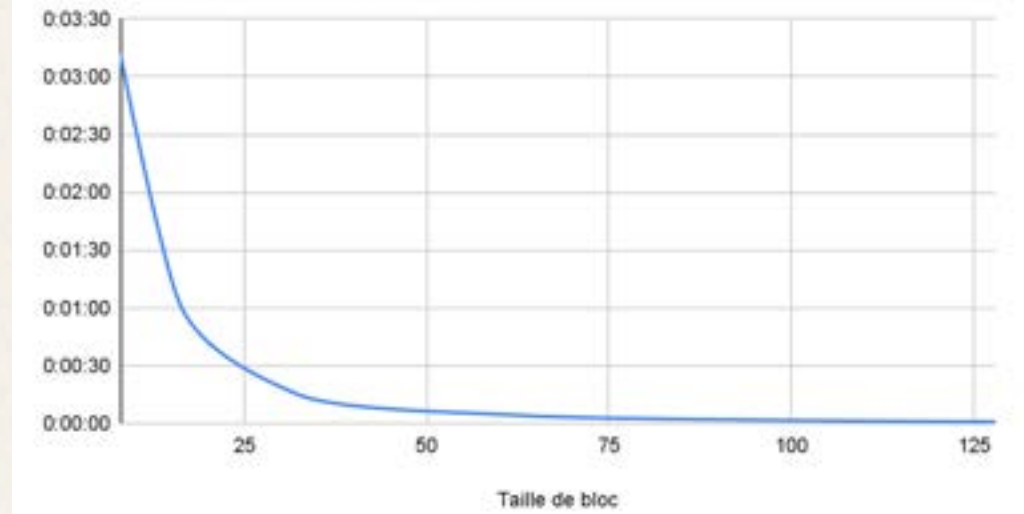
3 min 04



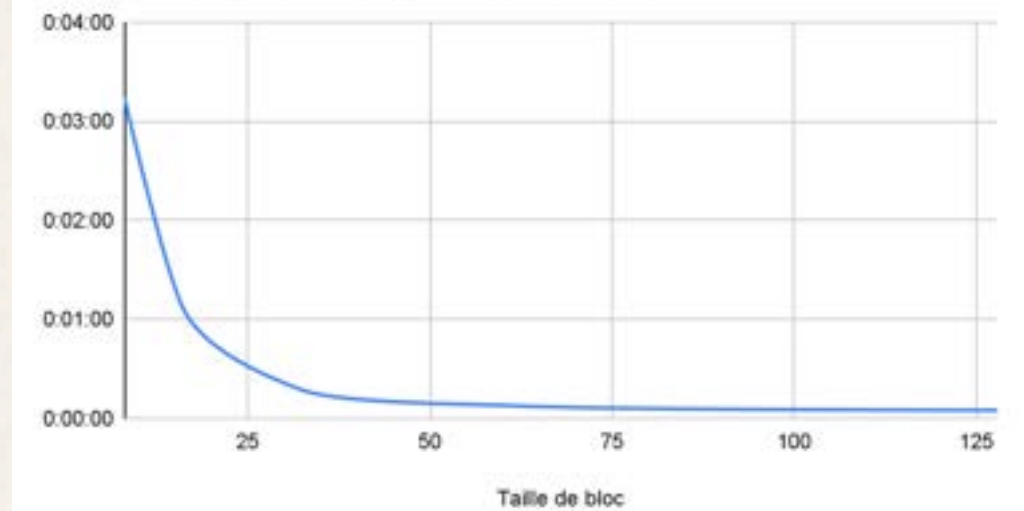
Chi-2

3 min 13

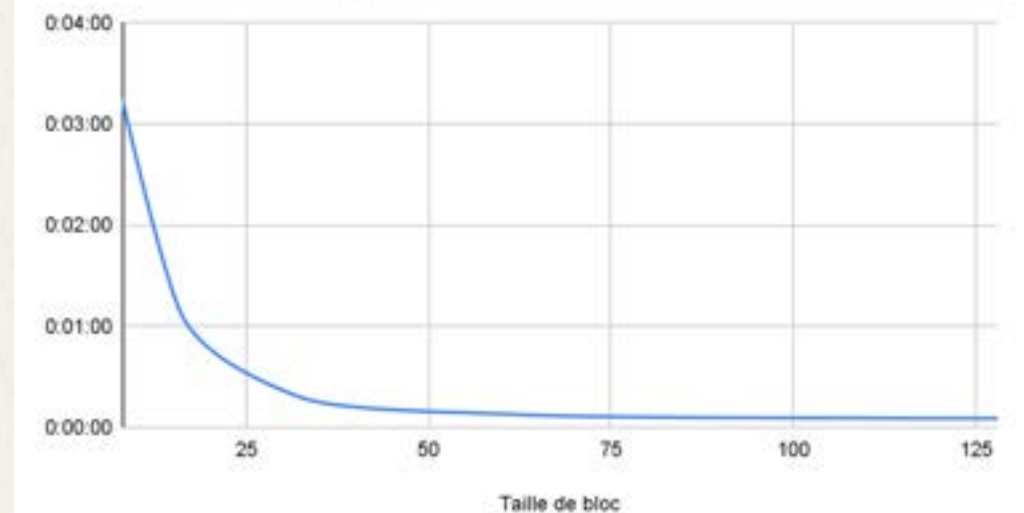
Temps de génération par rapport à la taille d'un bloc



Temps de génération par rapport à la taille d'un bloc



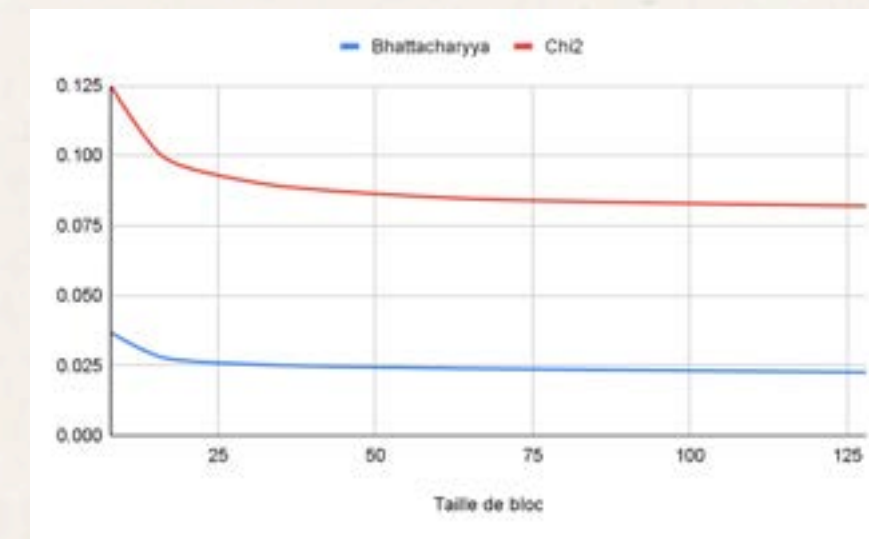
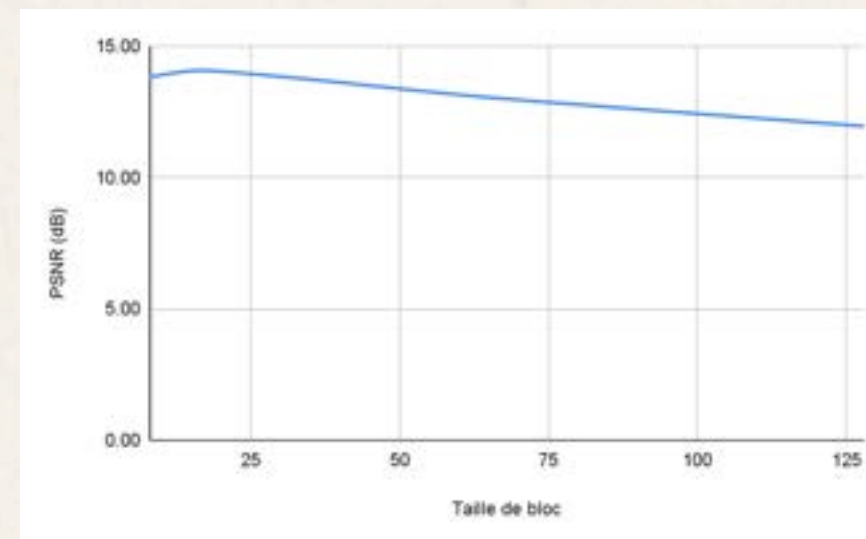
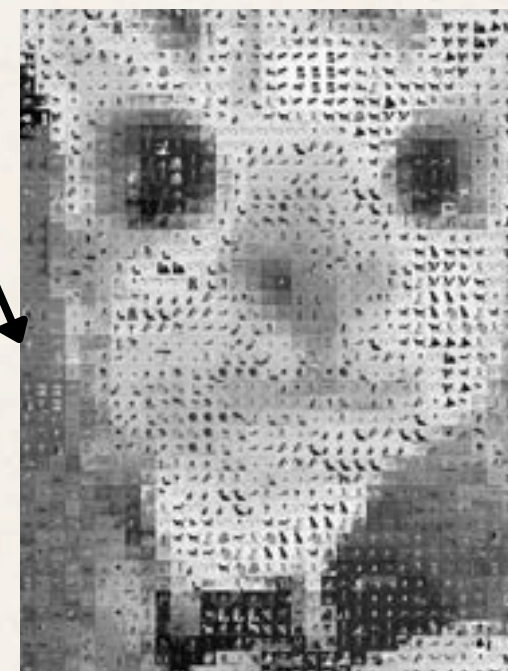
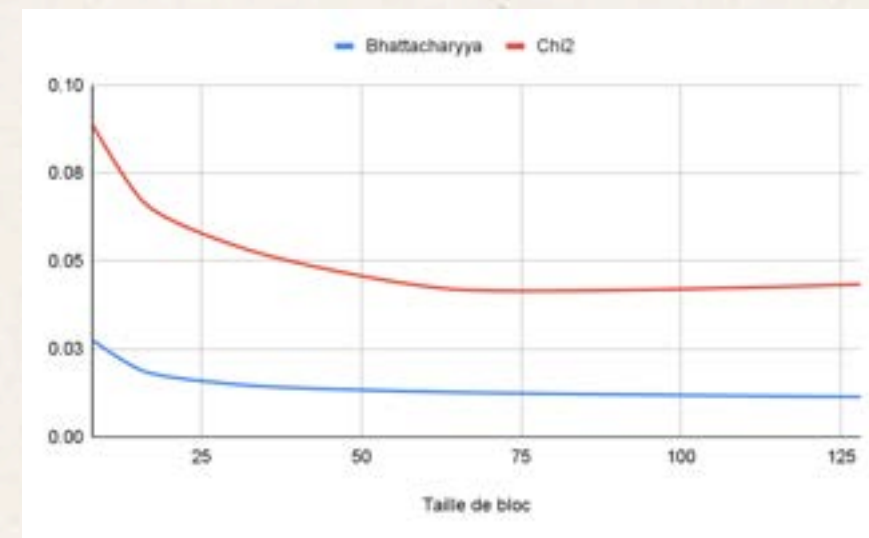
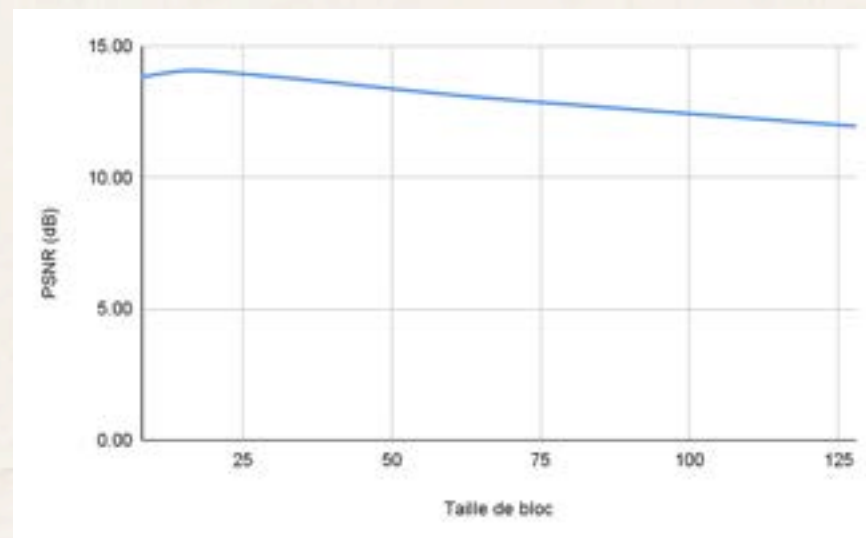
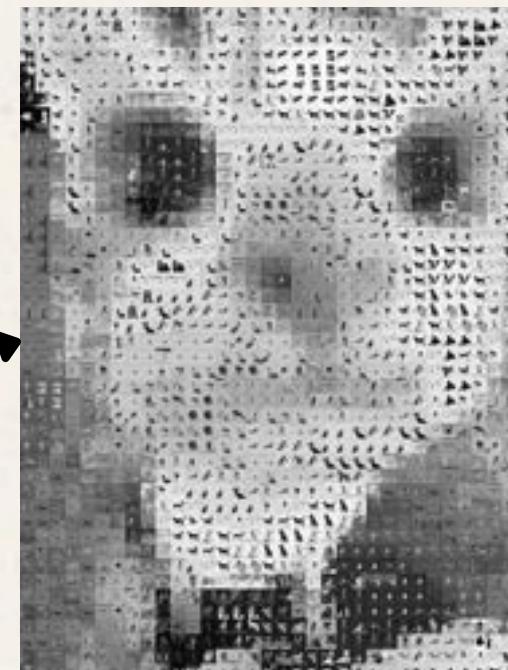
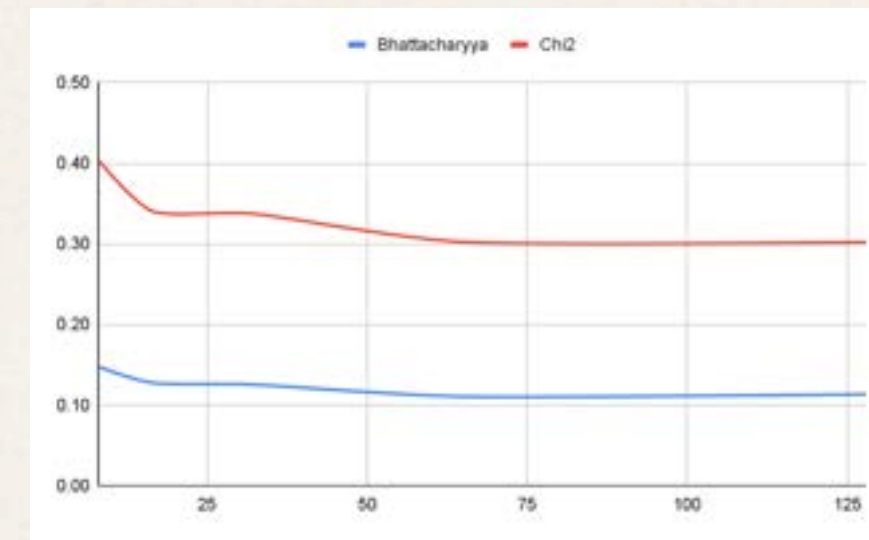
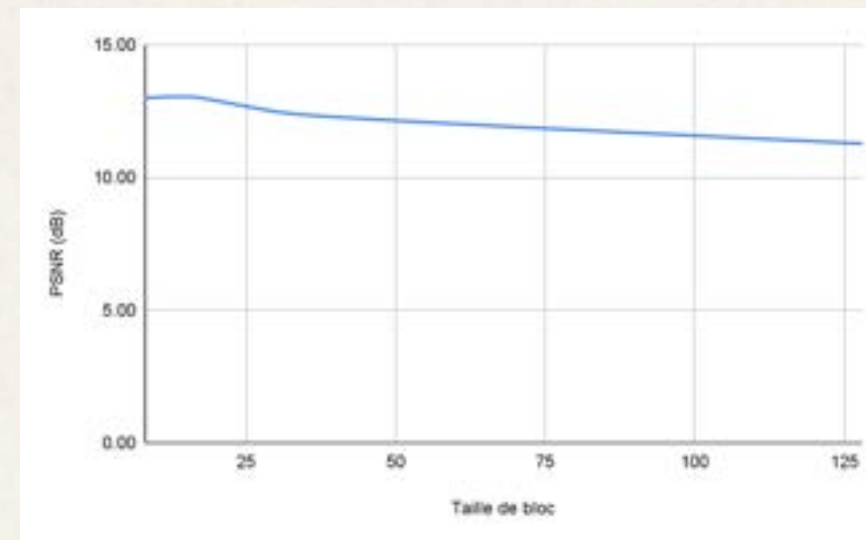
Taille de bloc par rapport à Temps



Résultats

Qualité de rendu (PPM)

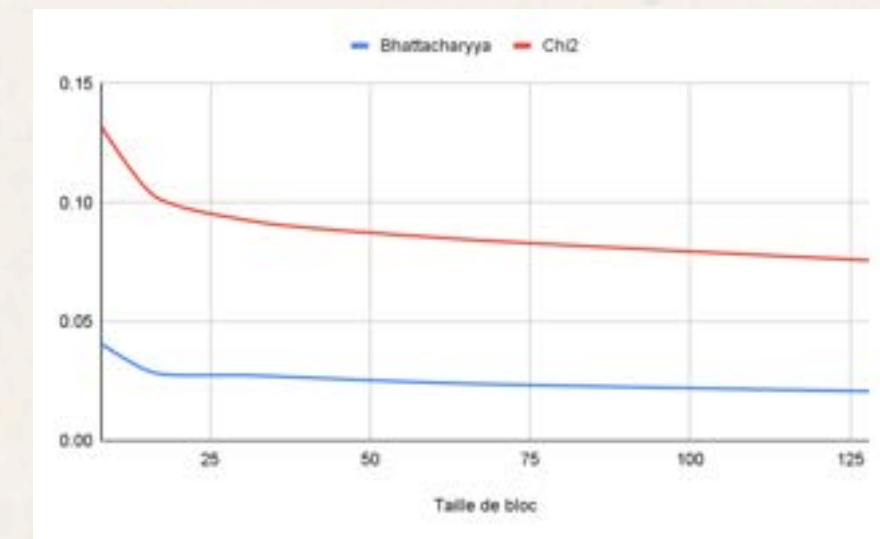
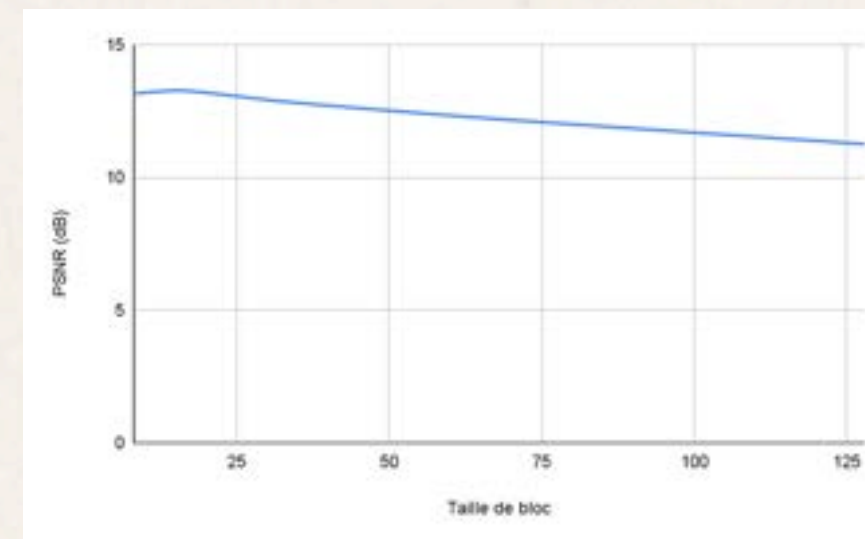
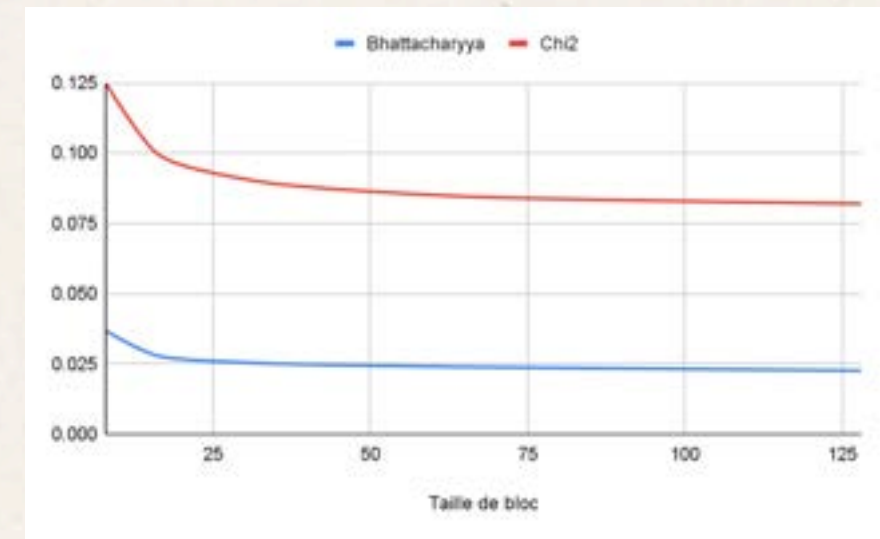
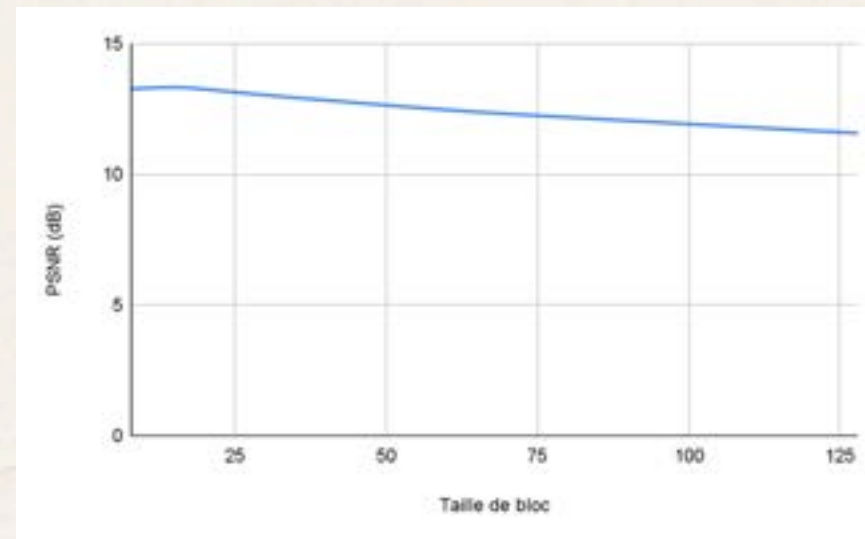
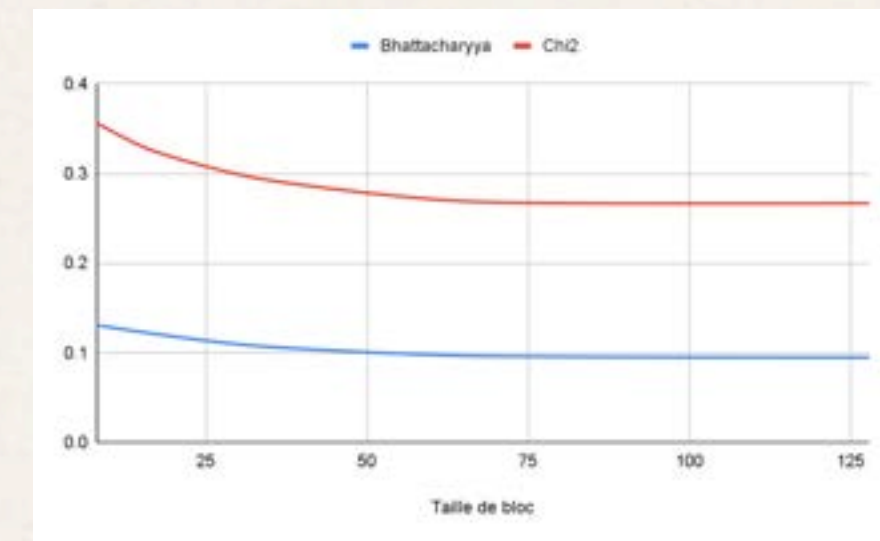
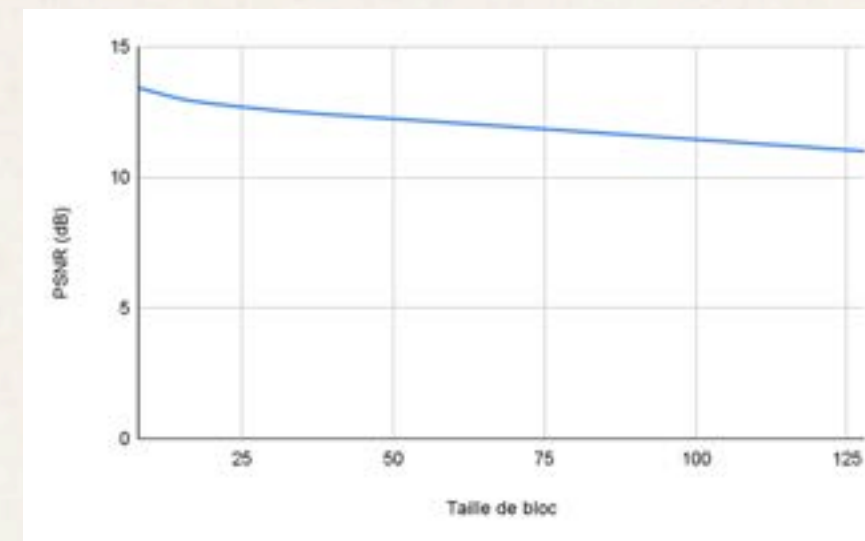
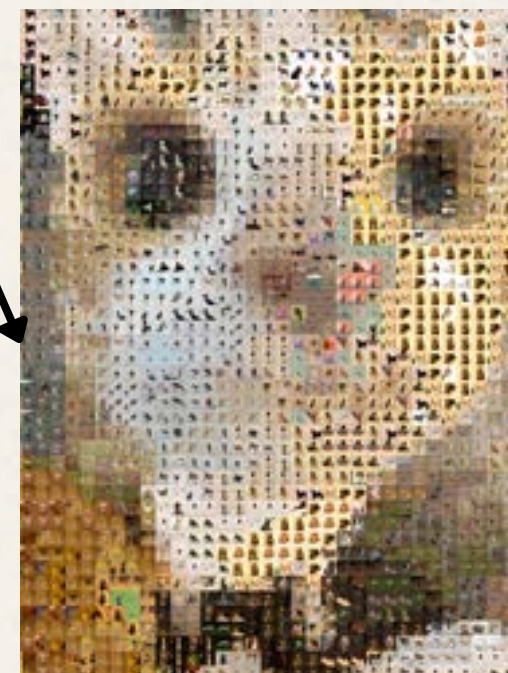
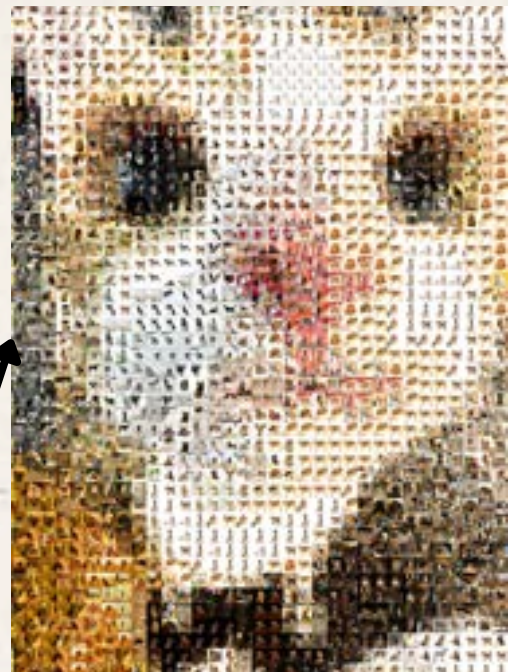
Blocs de 32 x 32



Résultats

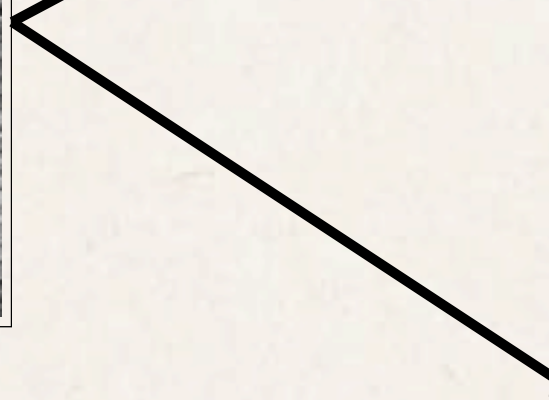
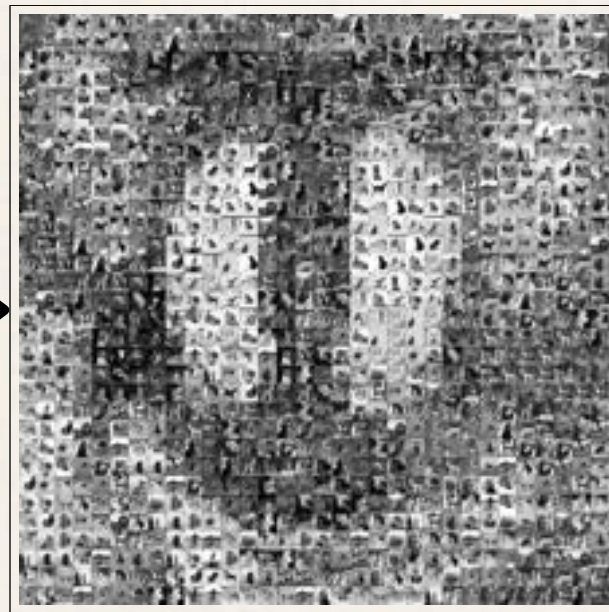
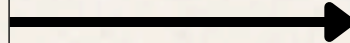
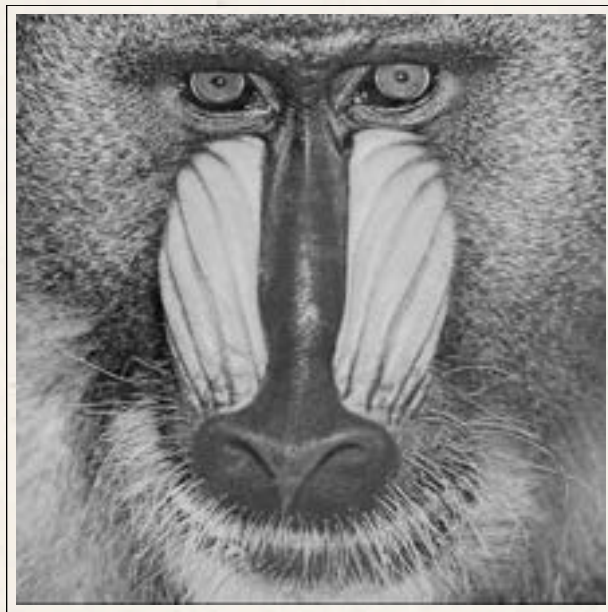
Qualité de rendu (PPM)

Blocs de 32 x 32

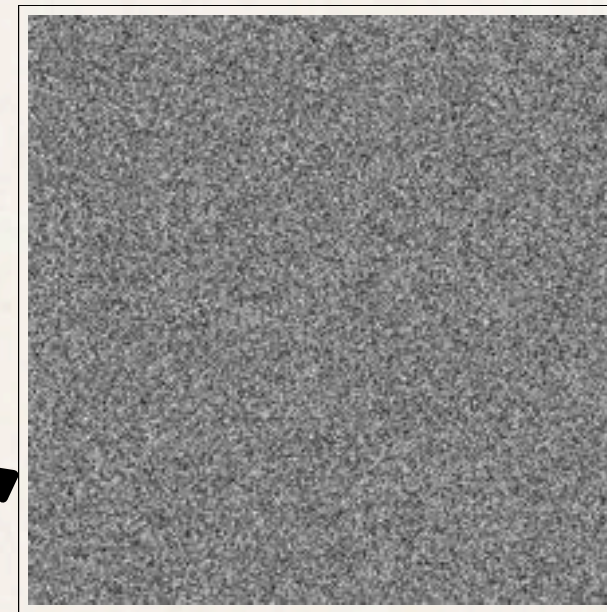


Résultats

Chiffrement et brouillage



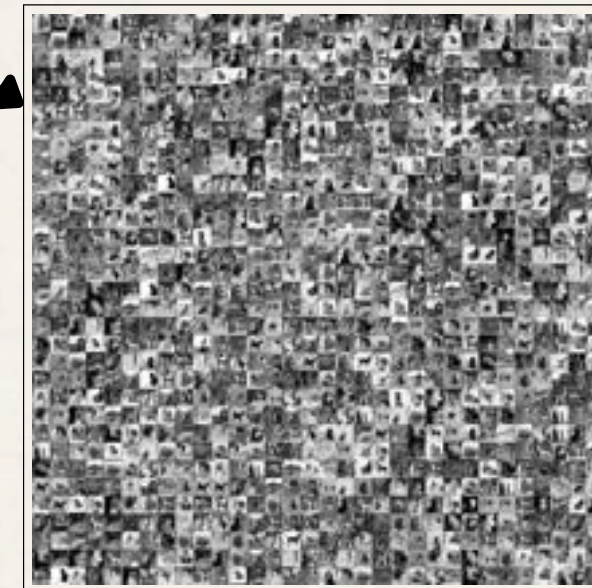
Vernam



PSNR de 8.1 dB



Permutation



PSNR de 8.6 dB



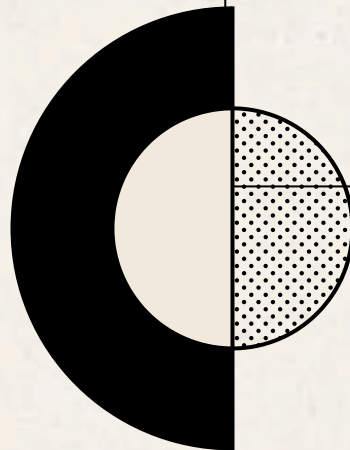
Demonstration



Conclusion

Bilan

Perspectives

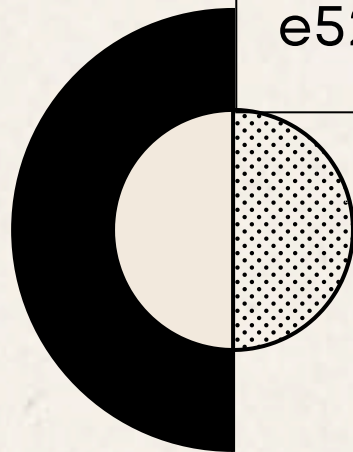


Ressources

<https://diglib.eg.org/server/api/core/bitstreams/9f0df882-5838-4067-a555-2ade20d925ca/content>

<https://dl.acm.org/doi/pdf/10.1145/298151.298213>

<https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=d9bba2c5949be015ba423682c15d37d6928abe52>



Merci beaucoup !

